THE UNITED REPUBLIC OF TANZANIA

# PRESIDENT'S OFFICE

## REGIONAL ADMINSTRATION AND LOCAL GOVERNMENT

## KASULU TOWN COUNCIL

## FORM FOUR PRE-MOCK EXAMINATION

**CODE: 041** **BASIC MATHEMATICS**

**Time: 3:00 Hours** **Tuesday, 20.05.2025**

**INSTRUCTIONS**

1.This paper consists of sections **A** and **B** with a total of fourteen (**14**) questions.

2.Answer ALL questions in sections **A** and **B**.

3.Each question in section **A** carries six (**06**) marks while each question in section **B** carries ten (**10**) marks

4. All working and answers for each question must be shown clearly.

5. Mathematical tables and non-programmable calculator may be used.

6.All communication devices and any unauthorized materials are **NOT** allowed in the examination room.

7.Write your **Examination Number** on every page of your answer sheet.

MsomiBora.com

SECTION A (60 Marks)

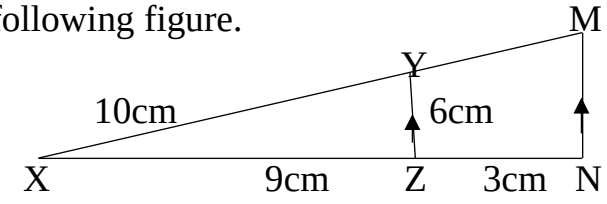
Answer **ALL** questions in this section.

1. (a) The numbers 28, 41, 42, 59, 70 belong to the set of natural numbers. By using these numbers.
   (i) Calculate the difference between the lowest common multiple of prime numbers and the greatest common factor of the remaining number.
   (ii) Express the answer obtained in 1.(a)(i) above in standard notation $A\times10^{n}$ convert into two significant figures where $1 \leq A < 10$ and n is integer

   (b) In a class of 40 students, 17 students are boys and the rest are girls. Determine. (i) the percentage of girls and boys.

   (ii) the number of boys in decimal.

2. (a) Express $\left(\frac{\sqrt{6}+\sqrt{3}}{\sqrt{6}-\sqrt{3}}\right) \times 3$ in the form $a + b\sqrt{c}$. Where $a, b$ and $c$ are integers.
   (b) Given a logarithmic equation $\log_3 x - \log_3(x-8) = 2$. Without using mathematical table or calculator, find the value of $x$.
3. (a) In a certain street of 200 houses, 170 have electricity and 145 have glass doors, how many houses have both electricity and glass doors, assuming that each house has either a glass door or electricity or both. Give out your answer by using Venn diagram.
   (b) The bag contains 4 red marbles ,6 green marbles and 5 yellow marbles. If two marbles taken random. What is the probability of taking a yellow and green marbles by using tree diagram? When marble ball is not replaced.
4. (a) Find the equation of the perpendicular bisector of a line joining the points (4,6) and (2,2).
   (b) Find the direction of the vector $\overrightarrow{PQ}$ whose initial point P is at (2,3) and end point Q is at (5,8).
5. (a)The area of a regular six – sided plot of a land inscribed in a circular track of radius r is 720 $m^2$ . Find the value of r by rounding off the answer to the nearest:
   (i) Tens.
   (ii) Tenths.
   (b) In the following figure.




   (i) Prove that the triangles XYZ and XMN are similar.

(ii) Calculate $\overline{MY}$ and $\overline{MN}$.

6. (a) Along the road going to school, Alex planted trees on each side of the road. If the road is $0.75km$ long and trees are $5m$ apart, how many trees are there?

(b) The time $t$ in seconds that mary takes to return home from school varies inversely with her average speed $v$ in metres per second. If Mary gets back home in half an hour at an average speed of 10 $m/s$

(i) Write an equation expressing $t$ in terms of $v$.

(ii) If her needs are to get back home in $(600 \div (-60 + 100))$ minutes, what will be her average speeds ?.

7. (a) The present ages of two friends are in the ratio 4: 3. If nine years to come, their ages will be in the ratio 23:18, find their present ages.

(b) Given the following transactions.

Sales for 2009 ……51,000/=
Stock at start …… 9,000/=
Purchases …… 34,650/=
Stock at close ……6,000/=
Returns on sales (inwards) ……1,000/=
Return outwards (return on purchase) ……150/=
Deduce: (i) Cost of sales.

(ii) Average stock.

(iii) Rate of stock turn.

(iv) Net sales(turnover).

8. (a) If the sum of the first $n$ terms of an arithmetic progression is $nP + \frac{1}{2}n(n-1)Q$,Where $P$ and $Q$ are constants, find the common difference.

(b) Find the rate percent per annum if 2,000 shillings amount to 2,662 shilling in $1\frac{1}{2}$ years, interest being compounded half – yearly?

9. (a) Given that $\alpha$ and $\beta$ are complementary angles and $\sin\alpha = \frac{5}{13}$, find $\tan\beta$. (b)A pole,1.6$m$ tall, stands on the top of a wall. From a point on the ground, the angle of elevation of the top of the pole is $60°$ and from the same point the angle of elevation of the top of the wall is $45°$. Find the height of the wall.

10. (a) What must be added to make the expression that the difference between the square of a number and twelve times the number will make it a perfect square.

(b) A train normally travels $240km$ at a certain speed. One day, due to bad weather, the train's speed is reduced by $20\,km/h$. So that the journey takes 2 hours longer. Find the normal speed of the train.

## SECTION B (40 marks).

Answer ALL questions in this section.

11. (a) The following table shows the distribution of marks scored by 42 candidates in mathematics exam at Utajili secondary school of pre-Mock 2025

| Marks (%) | 40-45 | 45-50 | 50-55 | 55-60 | 60-65 | 65-70 |
|---|---|---|---|---|---|---|
| Frequency | 7 | $x$ | 11 | $y$ | 4 | 2 |

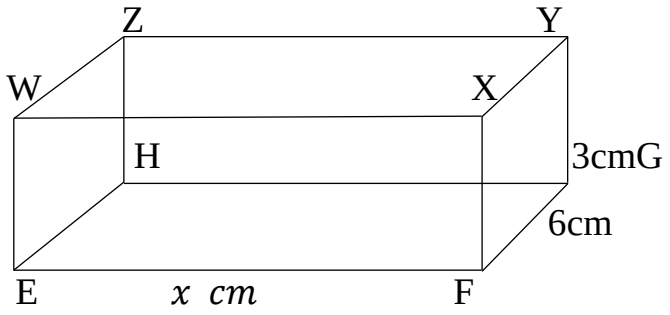
Calculate:

(i) The value of and if the Mode and the Modal class is 53.75 and 50-55 respectively.

(ii) The Mean (use an assumed Mean of the class mark within the Modal class).

(b) Two circles with centers A and B have radii $13cm$ and $10cm$, respectively and intersects at P and Q. If $\overline{PQ} = 12cm$, find the distance between the centres, giving your answer correct to 1 decimal place.

12. (a) The diagram below shows a rectangular block, $\overline{EF}$ = $x$cm , $\overline{FX}$=3cm and $\overline{FG}$ = 6cm.




(i) Find the value of $x$ .WHEN the total surface area of rectangular block is $180cm^2$.HENCE calculate The length of $\overline{EY}$.

(ii) The angle between $\overline{EY}$ and the base EFGH

(iii) A circular tin is placed in a square. If the side of the square is congruent to the diameter of the tin, calculate the percentage of the area of the square which remains uncovered.

(b) Find the distance in nautical miles between each of the following pairs of points:

(i) A$(29°N, 72°W)$ and B$(10°S, 72°W)$.

(ii) C$(30°N, 139°E$ and D$(45°N, 139°E)$.

13. (a) (i) Solve the simultaneous equations by using Determinant method.

$$\begin{cases} x - 3y = -24 \\ 5x - 2y = -29 \end{cases}$$

(ii) If $A = \begin{pmatrix} 2 & -3 \\ -2 & 1 \end{pmatrix}$ and $f(x) = x^2 - 3x - 4I$. Find $f(A)$, where $I$ is an identity matrix. What is the name of a such matrix?.

(b) A triangle AOB has vertices at O (0,0), A (2,1) and B (-1,3). If the Triangle is enlarged by enlargement factor K=2 and then translated by vector T(-3,-5).Find the vertices of the triangle.

(c) If farm measuring 72m by 88m is enlarged by a scale factor of 4,what will be the area of the enlarged farm?.

14. (a) During an inter-school quiz competition each correct answer $y$ was awarded number of points $x$ related by the function $y = 3x^2$.

(i) How many points would a competitor collect by answering 8 questions correctly?

(ii) Find the domain and range of the inverse of the given function.

(b) A company manufactures two types of products A and B are equipment in the factory allows for making at most $450$ of products A and $200$ of products B in one month. If the cost of A and B are $\epsilon 600$ and $\epsilon 900$ respectively, the profit per unit of A and B are $\epsilon 125$ and $\epsilon 200$ respectively. During the month of march $2007$,the company spent $\epsilon 360000$ to manufacture these products to make the greatest profit.

(i) How much of each type should be made in march $2007$?.

(ii) What will be the maximum profit produced?